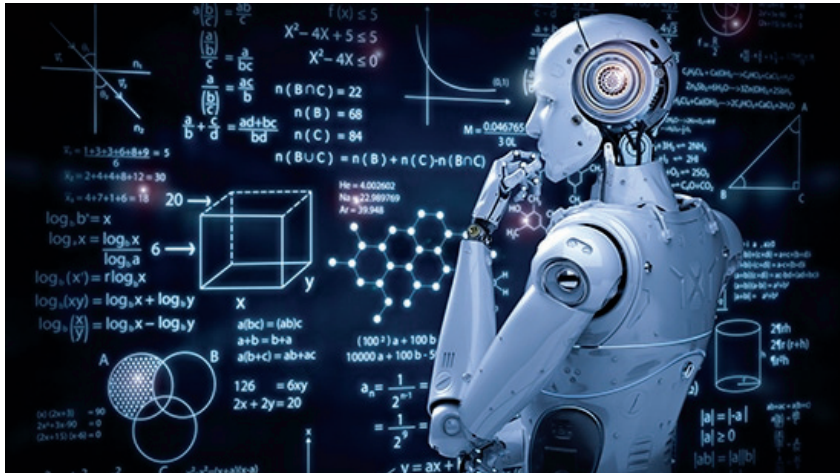


RISELab's SkyPilot helps optimise cloud costs



SkyPilot, an open source framework for running machine learning workloads on major cloud providers through a uniform interface, has been released by a group of researchers at the RISELab at UC Berkeley. The project's main objective is cost optimisation, which involves automatically locating the availability zone, area, and provider that is most affordable for the desired resources.

The system automatically decides which AWS, Azure, and Google Cloud regions have the resources (CPU/GPU/TPU) needed to run the project, as well as which are the most cheap. SkyPilot then completes three primary tasks: it controls job queueing and execution, synchronises user code and files to the destination, and provisions the cluster with automatic failover to other sites if there are capacity or quota difficulties.

SkyPilot is not RISELab's first open source initiative aimed at reducing cloud costs. The research centre earlier released SkyPlane, which speeds up and lowers the cost of transferring huge data sets between cloud providers.

Developing multi-cloud applications, utilising best-in-class hardware, and expanding the accessibility of limited resources like top-tier NVIDIA V100 or A100 GPUs, are a few advantages of SkyPilot.

The framework has two features: Managed Spot, a way to use less expensive spot instances with automatic pre-emption recovery, and Autostop, which helps automatically clear up unused clusters.

ClamAV finally releases version 1.0.0

Despite having a tiny changelog, ClamAV recently published its most current version, which can be regarded as historically significant. The achievement is ClamAV's arrival at version 1.0.0, which was released six months after the 20th anniversary of the software.

On May 8, 2002, ClamAV's first version (0.10) was released by the program's original developer, Tomasz Kojm. Following the previously published version 0.105.1, ClamAV 1.0.0 adds a useful new capability that allows users to scan and decrypt read-only, OLE2-based.xls (Excel) files that are password-protected by default. A revamped implementation of the all-match functionality, a better scanning function in archives, a remedy for compiler warnings, and numerous other bug fixes

Stability AI picks AWS as its go-to cloud provider

The open source artificial intelligence (AI) company Stability AI has chosen Amazon Web Services, Inc. (AWS), an Amazon.com, Inc. company, as its preferred cloud provider to build and scale its AI models for the creation of image, language, audio, video, and 3D content. Stability AI accelerates its work on open source generative AI models by utilising Amazon SageMaker (AWS's end-to-end machine learning service), as well as AWS's established compute and storage infrastructure. Additionally, Stability AI will work with AWS to make its open source tools and models accessible to worldwide startups, academics, and businesses.

Generative AI models from Stability AI can generate text, images, music, video, code, and more from straightforward language instructions. Models that are flexible to a range of tasks in areas like language, picture, audio, and video are known as generative AI or fundamental models. These models need a high-performance computing cluster with thousands of GPUs or AWS Trainium chips, extensive knowledge, and months of training. Stable Diffusion Version 2.0, which includes open source models for cutting-edge picture synthesis and enables users to produce new images from user-supplied text and image inputs, was recently released by the company.

When compared to self-managed machine learning infrastructure, Stability AI will save 58 per cent on training time and costs by using Amazon SageMaker to develop and test machine learning models. The business is leveraging Amazon Elastic Compute Cloud (Amazon EC2) to set up one of the biggest clusters of machine learning instances to run its open source models on AWS, making it simpler for developers to access those models.